U.S. Treasury Yield Curve Dynamics

Quant Mentorship Program

Spring 2026

Sahil Bhatia

Dataset and Source

For this project, I used daily U.S. Treasury yield data from the Federal Reserve Economic Data website, also called FRED. It is a free public database run by the St. Louis Federal Reserve. I downloaded ten different CSV files, one for each bond maturity: 1 month, 3 month, 6 month, 1 year, 2 year, 5 year, 7 year, 10 year, 20 year, and 30 year. The data covers roughly 2001 through early 2025, so I had about 6,000 trading days of data to work with.

In simple terms, a yield is the interest rate the government pays when it borrows money by selling bonds. For example, if a 10 year bond has a 4 percent yield, that means the government pays 4 percent per year for 10 years and then returns the original amount at the end. Usually, longer term bonds have higher yields because investors are giving up their money for a longer time. Sometimes that pattern reverses, though, and that is one of the main things this project looks at.

The data cleaning process was not too complicated. The main issue was missing values because the market is closed on weekends and holidays. To deal with that, I filled in gaps of up to 5 days by using the most recent available value. This let me handle normal market closures without adding unrealistic data. For the PCA part of the project, I removed any rows that still had missing values since PCA works best with a complete dataset.

Analysis Methods

I split the analysis into three main parts. The first part was exploratory data analysis. Before doing any modeling, I made four charts to better understand the dataset. The first chart shows all ten Treasury yields over time so it is easier to see the overall trends. The second shows the yield curve at four important points in history, which helps show how the curve changes under different economic conditions. The third tracks the gap between the 2 year and 10 year yield over time. This is known as the 2s10s spread and is often used as a warning sign for recessions. The fourth chart is a heatmap that shows how closely the different maturities move together from day to day.

The second part focused on the 2s10s spread. I calculated it by subtracting the 2 year yield from the 10 year yield for each day in the dataset. Most of the time this value is positive because longer term bonds usually have higher yields. When it becomes negative, that means the 2 year yield is higher than the 10 year yield, which is called a yield curve inversion. This has often happened before major U.S. recessions. In my graph, I marked inversion periods in red and recession periods in gray so it would be easy to compare them.

The third part was principal component analysis, or PCA. PCA is a method used to simplify a dataset when a lot of variables are closely related. Since I had 10 different yields that tend to move together, PCA helped reduce them into a smaller number of overall patterns that explain most of the changes in the data. Before running PCA, I standardized the data so that one maturity would not have more influence just because its values were larger. After that, I looked at

the results to see what each main pattern seemed to represent based on which maturities had the biggest effect on it.

Key Findings

One major finding was that the shape of the yield curve changes a lot depending on what is going on in the economy. I chose four dates that each matched a very different economic period. In early 2004, after the dot com crash, the Fed had lowered interest rates a lot to help the economy recover. Short term yields were very low, while long term yields stayed higher, so the curve had a steep upward slope. By mid 2006, the Fed had raised rates 17 times in a row, which pushed short term yields up so much that yields from 3 month to 30 year were all packed into a very small range of about 0.18 percent. That created an almost flat curve, which is usually seen as a warning sign. In March 2020 during COVID, the Fed cut rates back down to zero very quickly, so short term yields dropped hard. By mid 2023, after a series of aggressive rate hikes to fight inflation, the curve had become one of the deepest inversions in the dataset.

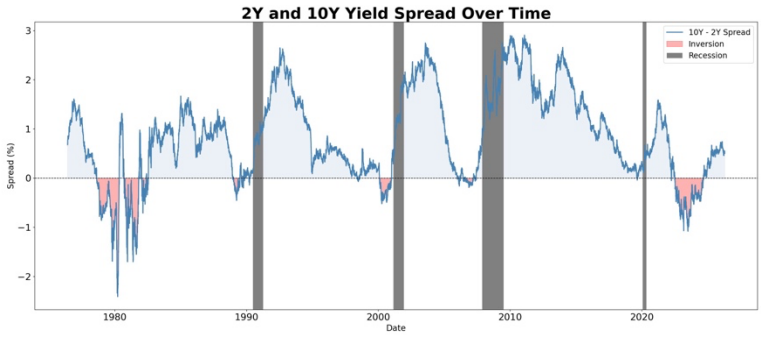
Another big finding was that every recession in the dataset was preceded by an inversion in the 2s10s spread. When I looked at the chart, the pattern stood out pretty clearly. Before the 2001 recession, the 2008 financial crisis, and the 2020 COVID recession, the spread had already gone negative. In each case, this happened about 12 to 18 months before the recession officially began. The most interesting case is the inversion in 2022 and 2023, since it was the deepest one in the whole dataset, but as of early 2025 there still had not been a clear recession. This could mean a recession is still delayed, or it could mean the yield curve is not as reliable as it was in the past.

I also found that most of the movement in the yield curve can be explained by just two main patterns. Even though I was working with 10 different yields, the PCA results showed that two components explained 98.77 percent of the total movement. The first component represents the overall level of the curve, meaning the whole curve moves up or down together. The second component represents the slope, which means the short end and long end move in opposite directions. That is what happens when the curve steepens or flattens. A third pattern related to curvature explained less than 1 percent of the variation. Overall, this shows that even though interest rate movements can look complicated at first, most of them come from just two main forces.

The last important result was that the math backed up what economists have already been saying for a long time. PCA was done using only the yield data, so it was not built around the 2s10s spread. Even so, when I compared the second principal component to the 2s10s spread, the correlation was 0.859, which is very high. So even without being told what to look for, the PCA

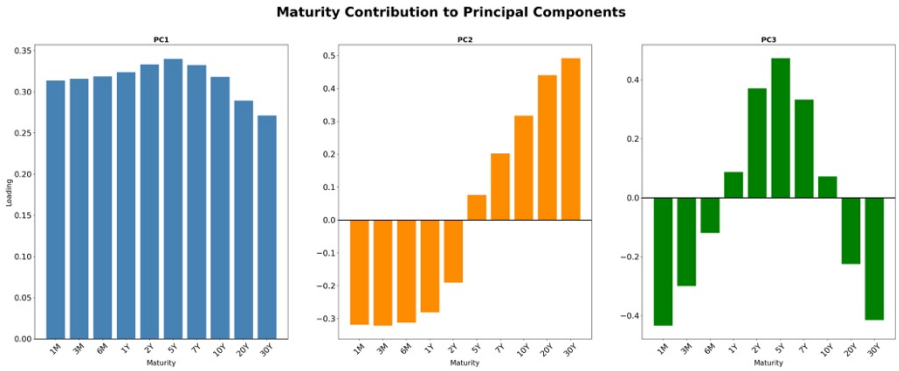
still found a pattern that was very close to the same recession signal economists have been using for years.

Figure 1 — 2Y vs. 10Y Yield Spread Over Time



Each gray band is a recession. Pink areas show when the curve was inverted (2Y yield > 10Y yield). Every recession was preceded by an inversion

Figure 2 — How Each Maturity Contributes to the Three PCA Factors



PC1 (Level): all bars point the same direction. PC2 (Slope): short maturities negative, long maturities positive. PC3 (Curvature): mixed pattern, explains less than 1% of movement

Reflection

Before this project, I did not know much about finance, so ideas like bonds, yields, and inversions were pretty new to me. That ended up helping because I had to really learn what the data meant instead of just running the code. One of the most interesting things I learned is that the yield curve says a lot about what investors think will happen in the economy.

The PCA results also surprised me. I expected all ten maturities to show different patterns, but almost all of the movement in the data, about 98.77 percent, was explained by just two main factors. That showed me how useful PCA can be for simplifying something that looks complicated at first.

The result that stood out most to me was the 2022 to 2023 inversion. Earlier in the data, every inversion was followed by a recession, but this time there was no clear recession by early 2025. That makes the signal harder to interpret and shows that data analysis does not always give a perfect answer.

If I kept working on this project, I would want to see whether the time between inversion and recession has been getting longer. I would also want to test whether adding other indicators, like unemployment or GDP growth, would make the signal more accurate.